

Topic Name :-Automatic Speech
Recognition Engine.

Team name :- TrailBlazers.

- Roles:-

Name	Roles
1) Komal Anil Mahajan.	Planner & Scrum master.
2) Ayush Ajit Baviskar.	Developer.
3) Mamata Kiran Chaudhari	Developer.
4) Bhagyashree Vijay Wagh	Developer.
5) Vishal Shivaji Gadekar	QA (Tester)
6) Srushti Subhash Mahajan	QA (Tester)

• Project Objectives

- Convert human speech into text accurately.
- Execute desktop automation using voice commands.
- Perform web automation through voice.
- Provide voice feedback after every command.
- Improve user productivity.
- Create a modular system that can easily support new commands in the future.

- Features :-

- User Authentication
- Voice Input & Speech Recognition
- Command Processing
- Desktop Automation
- Web Automation
- Information Services
- Voice Response
- Command History
- Voice Command Scheduling Module
- Voice Calculator Module
- Dashboard Module
- Email Sending Through Voice Command Module

1. User Authentication

Purpose:

Allow only authorized users to access the voice assistant.

Functional Requirements:

- The system should allow users to log in using a username and password.
- The system should verify the entered credentials.
- The system should allow access only if the credentials are valid.
- The system should display an error message for invalid credentials.
- The system should allow users to log out.
- The system should securely store user credentials (for a simple training project, this can be in a JSON file).

2.Voice Input & Speech Recognition

Purpose:-

Capture the user's voice and convert it into text.

Functional Requirements:-

- The system should capture voice through the microphone.
- The system should convert speech into text.
- The recognized text should immediately appear on the screen. The system should inform the user if no speech is detected.
- The system should handle speech recognition errors without crashing.

3.Command Processing

Purpose:-

Identify the user's command and execute the correct action.

Functional Requirements:-

- The system should compare the recognized text with predefined commands.
- The system should execute the correct action for valid commands.
- The system should display an error message for unsupported commands.
- The system should process only one command at a time.

4.Desktop Automation

Purpose:-

Allow users to open desktop applications using voice commands.

Functional Requirements :-

- The system should allow users to open:
- Calculator
- Notepad
- Google Chrome
- File Explorer
- The system should display a confirmation message after opening the application.

5.Web Automation

Purpose:-

Open websites and perform Google searches using voice commands.

Functional Requirements:-

- The system should:
- Open Google.
- Open YouTube.
- Search Google using voice.
- Display an error message if the internet connection is unavailable.

6.Information Services

Purpose :-

Provide basic information to the user.

Functional Requirements :-

The system should answer:

- What is the current time?
- What is today's date?
- The response should be displayed on the screen and spoken aloud.

7. Voice Response

Purpose :-

Allow the assistant to communicate with the user.

Functional Requirements :-

- The system should greet the user when the application starts.
- The system should speak confirmation after every successful command.
- The system should notify the user if a command fails.
- The system should say goodbye before closing the application.

8. Command History

Purpose :-

Store executed commands for future reference.

Functional Requirements:-

- The system should save every executed command in history.json.
- Each record should contain:-
 - Command Name
 - Date
 - Time
 - Status
- The system should allow users to view the command history.

9.Voice Command Scheduling Module

Purpose:-

Allow users to schedule voice commands to be executed automatically at a specified date and time.

Functional Requirements:-

- The system should allow users to schedule a voice command for a specific date and time.
- The system should allow users to choose the command to be executed.
- The system should save scheduled commands for future execution.
- The system should automatically execute the scheduled command at the specified time.
- The system should notify the user after successful execution.
- The system should allow users to view all scheduled commands.
- The system should allow users to edit scheduled commands.
- The system should allow users to delete scheduled commands.
- The system should display an error message for invalid date or time.

10. Voice Calculator Module

Purpose:

Perform mathematical calculations using voice commands.

Functional Requirements:

- The system should accept mathematical expressions through voice.
- The system should convert speech into text.
- The system should validate the mathematical expression.
- The system should evaluate the mathematical expression.
- The system should display the calculated result on the screen.
- The system should speak the calculated result.
- The system should maintain a history of previous calculations.
- The system should display an error message for invalid expressions.

11. Dashboard Module

Purpose:

Provide a centralized interface to monitor and manage the voice assistant.

Functional Requirements:

- The system should display the current user information.
- The system should display the last executed voice command.
- The system should display the command execution history.
- The system should display the total number of executed commands.
- The system should display scheduled commands.
- The system should display the current system status (Microphone and Internet).
- The system should provide quick access to all available modules.
- The system should refresh the displayed information automatically.
- The system should display notifications for successful and failed operations.

12.Email Sending Through Voice Command

Purpose:

Allow users to send emails using voice commands.

Functional Requirements:

- The system should allow users to specify the recipient's email address through voice.
- The system should allow users to specify the email subject through voice.
- The system should allow users to dictate the email message through voice.
- The system should convert the spoken content into text.
- The system should ask for confirmation before sending the email.
- The system should send the email to the specified recipient.
- The system should display a success message after sending the email.
- The system should display an error message if the email cannot be sent.
- The system should maintain a history of sent emails.

- Task Distribution to Developers.

Developers	Task 1	Task 2	Task 3	Task 4	Working Hours
1) Ayush Ajit Baviskar.	User Authentication (1 Hrs)	Voice Input & Speech Recognition (2 Hrs)	Web Automation (2 Hrs)	Voice Command Scheduling Module (3 Hrs)	Total : 8 Hrs Remaining : 0
2) Bhagyashree Vijay Wagh.	Command Processing (3 Hrs)	Desktop Automation (2 Hrs)	Email Sending Through Voice Command Module (2 Hrs)	Information Services (1 Hrs)	Total : 8 Hrs Remaining : 0
3) Mamata Kiran Chaudhari	Voice Response (2 Hrs)	Command History (1 Hrs)	Voice Calculator Module (2 Hrs)	Dashboard Module (3 Hrs)	Total : 8 Hrs Remaining : 0

- QA/Testers Task Distribution.

QA/Testers	Task 1	Task 2	Task 3	Task 4	Task 5	Task 6	Working Hrs
1) Srushti Subhash Mahajan	User Authentication (30 min)	Voice Input & Speech Recognition (1 Hrs)	Web Automation (1 Hrs)	Voice Command Scheduling Module (1 Hrs)	Email Sending Through Voice Command Module (1 Hrs)	Voice Calculator Module (30 min)	Total : 5 Hrs Remaining: 0
2) Vishal Shivaji Gadekar	Command Processing (1 Hrs)	Desktop Automation (30 min)	Voice Response (1 Hrs)	Command History (1 Hrs)	Information Services (30 min)	Dashboard Module (1 Hrs)	Total : 5 Hrs Remaining: 0

- # Acceptance Criteria

The project will be considered successful if:

- Supported commands execute successfully.
- Applications and websites open correctly.
- Voice responses are provided.
- Command history is stored in history.json.
- Speech is recognized correctly.
- The application handles invalid commands without crashing.